

Harim Choi

MACHINE LEARNING ENGINEER | PREDICTION · UNCERTAINTY · PRODUCTION ML

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Korean (native), English (professional working proficiency)

SUMMARY

I am a machine learning engineer interested in vectorizing reality, making predictions, and estimating uncertainty. Over seven years, I have worked with public-procurement, construction, and retail data, connecting tabular ML, NLP, computer vision, and LLM agents from data collection through evaluation and deployment. I quickly learn unfamiliar domains and inherited systems, then turn data, models, and operating workflows into working services. I build systems that balance predictive performance with reliability, cost, and operational constraints.

CORE CAPABILITIES

- **Rapid Problem Understanding & Execution** | Quickly learn unfamiliar domains and inherited systems, prioritize the highest-impact problems, and turn them into operable ML systems.
- **Model Diagnosis & Improvement** | Identify implementation errors and performance bottlenecks through experiments and analysis, then improve training, inference, and evaluation for the operating context.
- **Uncertainty-Aware Decision Support** | Go beyond point predictions by using distributions, intervals, calibration, and simulation to communicate reliability and risk.
- **Research-to-Production** | Adapt new research methods to real data, infrastructure, cost, and operating constraints, then implement them as repeatable pipelines.
- **End-to-End ML Delivery** | Connect data construction, training, evaluation, batch jobs, APIs, and service operations in a form that others can review, maintain, and extend.

TECHNICAL STACK

- **Tabular & Probabilistic ML** | TabPFN · TabICLv2 · Prior-Data Fitted Networks · Tabular ICL · AutoGluon · Quantile/Distributional Prediction · Conformal Prediction · Calibration · Monte Carlo Simulation
- **NLP & Document Intelligence** | Transformers · PEFT/LoRA · Weak Supervision · Knowledge Distillation · OCR · Document Parsing · Information Extraction
- **Vision & Multimodal Learning** | Vision-Language/Foundation Models · Object Detection · WSSS · Semantic Segmentation · 2D/3D Pose Estimation
- **RAG & Agentic Systems** | Vector Retrieval · Hybrid Search · Reranking · LangGraph · MCP · Tool Calling · HITL
- **Efficient Model Inference** | INT3/INT8 Quantization · Weight/KV/Activation Compression · ONNX CPU/GPU Inference
- **Production ML Systems** | Data/Training Pipelines · Parallel Inference · API/Batch Serving · Model/Artifact Lifecycle · Monitoring

EXPERIENCE

Sejong Bid Institute | Researcher

Apr 2026–Present

Public-procurement probabilistic ML, Document AI, and production operations

- **Probabilistic Bid ML & MLOps** | Took over a public-procurement AI system built by a five-person team over six months and rebuilt it in three months. Modeled regional competitor bids with quantile models and CQR/R2CCP, then generated about 80,000 candidate bids per notice from a q1000 model zoo using TablCLv2, CatBoost, XGBoost, and AutoGluon. Connected candidates to Monte Carlo simulation and operated the workflow through Airflow DAGs, parallel GPU inference, artifact checksums, dry-runs, and approval-gated deployment. The system serves more than 1,000 clients, improved the internal relative bid-award KPI by 30%, and reduced inference time from 136 seconds to 25 seconds.
- **Document AI OCR** | Collected HWP/HWPX and PDF notices, parsed them with rhwp, hwp5, and liteparse v2, and normalized the output to JSON. Used Gemini 2.5 Flash-Lite only to copy target fields while deterministic code validated amounts, ratios, and units. The pipeline processes about 2,000 new notices per day and achieved F1 0.985 on key-field extraction.

Hanmac Group | ML Engineer / Researcher

Nov 2024–Feb 2026

Procurement NLP, multimodal distribution forecasting, and RAG agents

- **Multimodal PQ Bid Prediction** | Divided PQ competition into eight contexts and modeled multimodal bid distributions with R2CCP. Corrected interval collapse in the public implementation with entropy regularization and per-bin conformal thresholds, preserving separated valid ranges before estimating bid probability with 500K Monte Carlo runs. Eight models trained on 69,934 cases reached 90.73% coverage on 13,984 time-ordered validation cases and improved the group-company PQ bid KPI by 35%.
- **Procurement NLP API** | Trained a RoBERTa-large+LoRA binary model for bid eligibility, then generated detailed-domain labels through a UMAP/HDBSCAN ontology and SBERT-RoBERTa teacher-based weak labeling. Quantized the binary and multiclass models to about 330 MB with static INT8 ONNX and deployed them through a FastAPI CPU batch path. The system reached Macro-F1 0.9639 and Accuracy 96.4% for eligibility, F1 0.90 for detailed domains, and about 50 ms inference per notice.
- **Cost-Estimation RAG Agent** | Built a FAISS-based vector index from government-document OCR output, standard-cost references, and unit-price data. Connected LangChain retrieval and a LangGraph workflow to cost estimation, quantity-takeoff sheets, and official-letter drafts.

Sanha General Technology | AI / Automation Engineer

Jan 2021–Nov 2024

Document, CAD, ERP, and drone-survey automation for construction engineering

- **Document Automation** | Structured heterogeneous construction documents with OCR and layout parsing, then added confidence-based review and rule-based post-processing.
- **CAD & ERP Automation** | Built an AutoCAD add-in and template automation for repetitive drawing work and connected document and drawing outputs to ERP input workflows.
- **Drone & Survey 3D Workflow** | Converted drone imagery and survey data into 3D outputs for field review.
- **Result** | Reduced document work from more than four hours to about 30 minutes and converted outsourced or manual document and drawing tasks into in-house automation tools.

Nielsen Korea GTC | Data Analyst

Aug 2019–Oct 2020

Computer vision and RPA for global retail product data

- **Retail Vision** | Extracted operational visual attributes from global retail product images and validated data quality.
- **Barcode Data** | Connected barcode-recognition results to the product database and built a unified image and product-information workflow.

- **RPA & Exception Routing** | Automated repetitive input and validation steps, routed only exceptions for review, and reduced department-wide processing time by about 40%.

EDUCATION

Chonnam National University | B.B.A.

Mar 2015–Jul 2018

PROJECTS & RESEARCH

WORK PROJECTS

Uncertainty-Aware Bid ML

Headline q1000 and Monte Carlo decision ML for bid probability and uncertainty

Stack Python · Airflow · DAG · TabICLv2 · CatBoost · XGBoost · AutoGluon · CQR · R2CCP · Monte Carlo Simulation · GPU Inference

- **Why · How** | A single point-prediction model could not provide both bid probability and uncertainty across different competitive environments. Modeled regional competitor bids with XGBoost+CQR and R2CCP, then evaluated q1000 model-zoo candidates with Monte Carlo simulation.
- **Result** | Improved the bid KPI by 30% over the previous single-quantile model, reduced inference from 136 seconds to 25 seconds, and delivered client-level risk and bid-probability reports.

Document AI OCR

Headline HWP/PDF procurement-document extraction · F1 0.985 · 2,000 documents/day

Stack Python · HWP/HWPX · PDF · ZIP · rhwp · hwp5 · liteparse v2 · Gemini 2.5 Flash-Lite · JSON · ThreadPool

- **Why · How** | Core features such as ownership shares and capability criteria existed only in notice bodies. Collected HWP/HWPX and PDFs, normalized them to JSON, and separated LLM field copying from deterministic validation of amounts, ratios, and units.
- **Result** | Automatically processed about 2,000 notices per day and reached F1 0.985 on key-field extraction.

Procurement NLP

Headline CPU NLP API that classifies bid eligibility and detailed work domains from titles

Stack Python · PyTorch · KLUE RoBERTa-large · LoRA/PEFT · SBERT · UMAP · HDBSCAN · ONNX Runtime INT8 · FastAPI

github.com/HarimxChoi/nlp-analysis-agent

- **Why · How** | Needed immediate decisions from titles despite missing fine-grained labels and no inference GPU. Combined RoBERTa-large+LoRA, UMAP/HDBSCAN ontology construction, and SBERT-RoBERTa weak labeling, then deployed INT8 ONNX models through a FastAPI batch API.
- **Result** | The bid-eligibility model achieved Macro-F1 0.9639; detailed-domain classification achieved F1 0.90; both ran at about 50 ms per notice on CPU.

R2CCP Bid Prediction

Headline Multimodal PQ bid distributions across eight competitive contexts

Stack Python · PyTorch · R2CCP · Per-bin Conformal Threshold · Entropy Regularization · Monte Carlo Simulation

github.com/HarimxChoi/ensemble-bid-prediction

- **Why · How** | PQ scoring creates separated valid bid ranges and multimodal bid behavior. Modeled eight competitive contexts with R2CCP, then prevented interval collapse using entropy regularization and per-bin conformal thresholds before estimating bid probability with 500K Monte Carlo runs.
- **Result** | Trained eight context models on 69,934 cases, reached 90.73% coverage on 13,984 time-ordered validation cases, and improved group-company PQ bid KPI by 35% in operation.

MODEL EFFICIENCY & AGENT SYSTEMS

WarpQuant

Headline INT3 LLM quantization across weights, KV cache, and activations

Stack Python · PyTorch · Hadamard Rotation · Block-GPTQ · Output-Fisher · TurboQuant KV Cache · GGUF

github.com/HarimxChoi/WarpQuant

- **Why · How** | TurboQuant showed that Hadamard rotation could suppress KV-cache outliers. Applied the same idea to weights with block-GPTQ and Output-Fisher restoration, then added TurboQuant KV cache and token-wise activation quantization.
- **Result** | Compressed the Qwen3.8-27B text backbone to 3.6165 bpw and 11.32 GiB. It retained about 99% of BF16 performance on MMLU and common-sense evaluations; ARC rose from 52.1% to 56.8%, while PPL increased by 7.5%.

google-surf-mcp

Headline Integrated MCP for agent search, web extraction, and academic PDF parsing

Stack TypeScript · MCP · Google Search · Google Scholar · Playwright · Readability · Spatial PDF Parsing · SSRF Protection · Vitest

github.com/HarimxChoi/google-surf-mcp

- **Why · How** | Agents needed reliable access to web pages and academic PDFs beyond search snippets. Unified Google Search, Google Scholar, parallel search, Playwright/Readability extraction, spatial PDF parsing, provider fallbacks, and CAPTCHA handoff in one MCP.
- **Result** | Enabled agents to handle general search, academic search, web extraction, and PDF parsing through one MCP. Published the npm package after 388 search, extraction, recovery, and security tests; it reached the MCP TOPLIST Top 1%.

Monogram

Headline Personal knowledge management from one share action to structured, searchable knowledge

Stack Python · 5-stage LLM Pipeline · Atomic Git Tree · Markdown · Telegram · GitHub API · MCP · Dashboard

github.com/HarimxChoi/monogram

- **Why · How** | Bookmarks across news, social media, arXiv, YouTube, and reports lost context and became difficult to retrieve. Unified links, messages, documents, and code in one input path; a five-stage LLM pipeline structures content, topics, sources, and related knowledge as Markdown, while atomic Git commits preserve provenance.
- **Result** | Implemented a share-to-knowledge workflow with searchable dashboard access, versioned sources, and 13 MCP tools that reuse the same knowledge as agent context.

Bau Browser

Headline Agentic browser with user-controlled privacy and execution authority

Stack TypeScript · Electron · MCP · SQLite · Qwen3.5-4B · XGrammar · TaskSpec · ActionScope · DecisionLog

- **Why · How** | Built a local-first Electron browser that keeps browsing data, agent memory, and execution records locally. Agents propose work first; they create arguments only within approved sites and action scopes, and state-changing actions require approval and produce a receipt and DecisionLog.
- **Result** | Built a desktop MVP with native privacy controls, action approval, and execution logs. Every state-changing action follows proposal, approval, execution, and receipt; the system generated the exact action and arguments for 36/36 new-domain tasks, with zero out-of-scope actions.

LangGraph Travel Agent

Headline LangGraph travel agent that produces three bookable packages from a natural-language request

Stack Python · LangGraph · FastAPI · AsyncIO · InMemorySaver · Amadeus API · Hotelbeds API · HubSpot · Twilio

github.com/HarimxChoi/langgraph-travel-agent

- **Why · How** | For a US travel agency, structured origin, destination, dates, budget, and preferences; concurrently gathered flights, hotels, places, and restaurants; then assembled Budget, Balanced, and Premium packages with consultant review and CRM follow-up.
- **Result** | Implemented the workflow from request intake through package generation, consultant review, CRM storage, and customer contact. Delivered to a US travel agency; the IP was later transferred under contract and released as public OSS.

RESEARCH & DECISION SYSTEMS

Weakly Supervised Semantic Segmentation

Headline Commissioned WSSS SOTA research · COCO-Val mIoU 53.31% · +1.5pp vs. WeCLIP+

Stack Python · PyTorch · CLIP ViT-B/16 · DINOv2 · WeCLIP+ · RFM · COCO-Val

github.com/HarimxChoi/wsss-refined-pseudolabels

- **Why · How** | In a commissioned research project, the client asked for a WSSS method that improved on the then-SOTA WeCLIP+. We kept CLIP and DINOv2 frozen, trained only the fusion head and decoder, repaired pixels where their representations disagreed, and reused the refined pseudo-masks for self-training.
- **Result** | Achieved mIoU 53.31% on all 40,137 COCO-Val 2014 images, 1.5 percentage points above WeCLIP+ ViT-B/16 at 51.8%.

Vargo

Headline Can the best LLM-agent configuration be predicted before execution?

Stack Python · ALFWorld · 134×29 Evaluation Matrix · Router · Bandit · Activation Probe · Verifier · Sealed-cell Evaluation

- **Why · How** | Ran the same 134 ALFWorld tasks across 29 agent configurations and compared routers based on task text, embeddings, hidden states, prior results, and bandit signals. Then compared pre-execution selection with executing several candidates and verifying outcomes.
- **Result** | Pre-execution signals could not reliably identify each task's best configuration. Verified execution increased recovered oracle gap from 54% to 85% and recovered 100% within an 11-configuration deployable pool. The resulting paper is under review at TMLR.

EMH Agent

Headline Cost-aware reinforcement-learning investment agent using point-in-time data

Stack Python · TQC · IVV/IEF/SHV · Point-in-time Panel · Information Agent · Transaction Costs · Drawdown

github.com/HarimxChoi/emh-agent

- **Why · How** | Separated real-time information collection from decision making so the TQC policy used only information available at each date. Trained a three-seed ensemble to allocate IVV, IEF, and SHV, then evaluated it after transaction costs, slippage, and borrowing costs.
- **Result** | Across 756 trading days from 2019 to 2021, the cost-aware backtest returned 183.9% versus 99.6% for IVV. Maximum drawdown was -14.2% versus -33.8% for IVV.

MyShot

Headline 3D golf-swing reconstruction and coaching from phone video

Stack Python · PyTorch · RTMPose · MotionAGFormer · GolfPose · GVHMR · Vicon · GolfDB · CMU MoCap

- **Why · How |** Extracted 2D joints with RTMPose, used MotionAGFormer to reconstruct 3D poses, trained on Vicon motion and pseudo-3D golf video data, and computed X-Factor, knee angle, and body rotation across swing phases.
- **Result |** Recorded MPJPE 35.6 mm and X-Factor MAE 3.1° on Clean-2D evaluation; predicted X-Factor changes correlated with real motion-capture data at mean $|r|$ 0.95 across six trials.

E-AT

Headline ISIC skin-lesion classification · minimum ECE 1.03% vs. CR-SAM 1.7% (AAAI 2024)

Stack Python · PyTorch · ConvNeXtV2-Tiny · Focal Loss · R-Drop · FGM · ISIC 2019 · Reliability Diagram

- **Why · How |** Combined focal loss, R-Drop, and FGM with ConvNeXtV2-Tiny on the balanced four-class ISIC 2019 dataset, tracking Macro-F1 and 15-bin ECE together by epoch.
- **Result |** Reached peak validation Macro-F1 0.8647. The minimum-ECE checkpoint recorded 1.03%, 0.67 percentage points below the then-SOTA CR-SAM's 1.7% (AAAI 2024), then evaluated classification and confidence calibration within one training framework.